

Unbounded Magnetic Ground-State Multiplicity by Products of Half-Flux Annuli

Candidate full-resolution packet for arXiv:math/0205088

12 August 2026

Status. Candidate full resolution of the final open problem in Lieb–Seiringer–Yngvason [1], subject to human validity and novelty review.

1 Source question

The source is E. H. Lieb, R. Seiringer, and J. Yngvason, *Poincaré inequalities in punctured domains*, arXiv:math/0205088, later published in *Annals of Mathematics* 158 (2003), 1067–1080 [1]. On printed page 1079, its final open problem asks:

The dimension of the ground state manifold $\mathcal{M}_A^{p,q}$ can be bigger than one. In the appendix we give an example for the case $p = q = 2$ where its dimension is two. It would be interesting to construct examples (or prove their existence) where the dimension can be arbitrarily large.

- Since Theorem 3 is proved by a compactness argument, the constant C appearing in (7) is left unspecified. It would be desirable to obtain a decent upper bound for this value. Also the optimal values of the constants $S_\delta^{p,q}$ appearing in Theorem 2 are in general unknown. Indeed, decent estimates of $S_{p,q}$ in Theorem 1 are not readily available when $p, q \neq 2$.
- The dimension of the ground state manifold $\mathcal{M}_A^{p,q}$ can be bigger than one. In the appendix we give an example for the case $p = q = 2$ where its dimension is two. It would be interesting to construct examples (or prove their existence) where the dimension can be arbitrarily large.

Appendix: Spectrum of $-(\nabla + iA)^2$

Here we prove two facts about $-(\nabla + iA)^2$ which were used in the text. As before, we have a bounded, connected domain Ω in $\mathbb{R}^n$ with the cone property. Connectedness is not really necessary here, but the number of connected components should be finite. The vector field A is bounded and measurable. The boundedness is not crucial but we assume it for simplicity.

We define the eigenvalues (spectrum) E_k and eigenfunctions ϕ_k of $-(\nabla + iA)^2$ in $L^2(\Omega)$ by means of quadratic forms as in [1], i.e., E_{k+1} is defined by

$$(39) \quad E_{k+1} = \inf \left\{ \|(\nabla + iA)f\|_{L^2(\Omega)}^2 : \|f\|_{L^2(\Omega)}^2 = 1, \int_{\Omega} f \overline{\phi_j} = 0 \text{ for } j = 1, \dots, k \right\}.$$

Then, by standard methods (using the Rellich-Kondrashov theorem), one shows that there is a minimizer for E_{k+1} , which is called ϕ_{k+1} . In the text, $\sqrt{E_1}$ was called $E_A^{2,2}$.

The two facts are the following.

LEMMA 2 (Spectrum of $-(\nabla + iA)^2$).

- The spectrum is discrete; i.e., the number of eigenvalues less than any number E is finite.*
- The multiplicity of E_1 can be greater than one.*

Proof. To prove A we suppose that there are infinitely many eigenvalues below E . If $\psi \in W^{1,2}(\Omega)$, with $\|\psi\|_{L^2(\Omega)} = 1$ and $\|(\nabla + iA)\psi\|_{L^2(\Omega)} \leq E^{1/2}$, then, since A is bounded, $\|\nabla\psi\|_{L^2(\Omega)} \leq E^{1/2} + \|A\|_{L^\infty(\Omega)}$. Thus, the infinite sequence of functions $\phi_1, \phi_2, \dots$ is bounded in $W^{1,2}(\Omega)$. By the Rellich-Kondrashov theorem this sequence has a subsequence that converges strongly in $L^2(\Omega)$. This is impossible since the ϕ_i 's are orthonormal (and hence $\|\phi_i - \phi_j\|_{L^2(\Omega)} = \sqrt{2}$ for $i \neq j$).

For $p = q = 2$, the source identifies $\mathcal{M}_A^{2,2}$ with the ground eigenspace of the Neumann realization of

$$H_A = -(\nabla + iA)^2$$

on a bounded connected domain with the cone property. Equivalently, it is the minimizer space for

$$E_A^{2,2} = \inf_{0 \neq u \in H^1(\Omega)} \frac{\|(\nabla + iA)u\|_{L^2(\Omega)}}{\|u\|_{L^2(\Omega)}}.$$

2 Proof intuition

The source already supplies the elementary multiplicity-two building block: an annulus carrying half a quantum of Aharonov–Bohm flux. Angular Fourier modes then see the shifted integers $\ell - \frac{1}{2}$, so precisely the two modes $\ell = 0, 1$ minimize the radial energy. The upgrade is to take Cartesian products. Magnetic Neumann operators on a product add, their ground energies add, and their ground eigenspaces tensor. Thus k half-flux annuli have ground multiplicity 2^k .

3 Formal result

Theorem 1. *For every integer $k \geq 1$, there are a bounded connected open set $\Omega_k \subset \mathbb{R}^{2k}$ with the cone property and a vector field $A_k \in C^\infty(\overline{\Omega_k}; \mathbb{R}^{2k})$ such that*

$$\dim \mathcal{M}_{A_k}^{2,2} = 2^k.$$

Consequently, the dimensions of the ground-state manifolds occurring in the setup of [1] are arbitrarily large.

Proof. Fix $0 < a < b < \infty$, and let

$$D = \{x = (x_1, x_2) \in \mathbb{R}^2 : a < |x| < b\}, \quad A_0(x) = \frac{(-x_2, x_1)}{2|x|^2}.$$

Thus, in polar coordinates, $A_0 = (2r)^{-1}e_\theta$. It is smooth and bounded on $\overline{D}$. Let H_0 be the self-adjoint magnetic Neumann operator associated with the closed quadratic form

$$q_0[u] = \int_D |(\nabla + iA_0)u|^2 dx, \quad \text{dom } q_0 = H^1(D).$$

Expand u in angular modes as

$$u(r, \theta) = \sum_{\ell \in \mathbb{Z}} e^{-i\ell\theta} g_\ell(r).$$

Orthogonality gives

$$q_0[u] = 2\pi \sum_{\ell \in \mathbb{Z}} \int_a^b \left(|g'_\ell(r)|^2 + \frac{(\ell - \frac{1}{2})^2}{r^2} |g_\ell(r)|^2 \right) r dr. \quad (1)$$

For $\nu \geq 0$, write

$$\mu(\nu) = \inf_{0 \neq g \in H^1(a,b)} \frac{\int_a^b \left(|g'|^2 + \frac{\nu^2}{r^2} |g|^2 \right) r dr}{\int_a^b |g|^2 r dr}.$$

If $0 \leq \nu_1 < \nu_2$, then for every nonzero g ,

$$\frac{Q_{\nu_2}[g]}{\|g\|_{L^2(r dr)}^2} \geq \frac{Q_{\nu_1}[g]}{\|g\|_{L^2(r dr)}^2} \frac{\nu_2^2 - \nu_1^2}{b^2}.$$

Taking infima shows that $\mu(\nu)$ is strictly increasing:

$$\mu(\nu_2) \geq \mu(\nu_1) + \frac{\nu_2^2 - \nu_1^2}{b^2} > \mu(\nu_1). \quad (2)$$

The smallest possible value of $|\ell - \frac{1}{2}|$ is $\frac{1}{2}$, attained exactly for $\ell = 0, 1$. Equations (1)–(2) therefore show that the bottom of $\sigma(H_0)$ occurs in exactly these two angular sectors. In each sector, the lowest radial eigenvalue is simple by the standard one-dimensional regular Sturm–Liouville theorem. Hence

$$\dim \ker(H_0 - \mu(\frac{1}{2})) = 2. \quad (3)$$

Now set

$$\Omega_k = D^k \subset (\mathbb{R}^2)^k = \mathbb{R}^{2k}$$

and, writing $x = (x^{(1)}, \dots, x^{(k)})$, define

$$A_k(x) = (A_0(x^{(1)}), \dots, A_0(x^{(k)})).$$

The domain Ω_k is bounded, connected, and has the cone property (indeed it is a bounded Lipschitz product domain), while A_k is smooth and bounded on its closure.

The magnetic form on Ω_k is the sum of the forms acting in the individual variables. Thus its Neumann operator is the tensor sum

$$H_k = \sum_{j=1}^k I \otimes \dots \otimes I \otimes H_0 \otimes I \otimes \dots \otimes I.$$

Because H_0 has compact resolvent, an orthonormal eigenbasis for H_0 produces, by tensor products, an orthonormal eigenbasis for H_k . The corresponding eigenvalues are all sums $\lambda_{n_1} + \dots + \lambda_{n_k}$. Their least value is $k\mu(\frac{1}{2})$, and equality holds exactly when every tensor factor lies in the two-dimensional space in (3). Therefore

$$\ker(H_k - k\mu(\frac{1}{2})) = \ker(H_0 - \mu(\frac{1}{2}))^{\otimes k}$$

and

$$\dim \mathcal{M}_{A_k}^{2,2} = \dim \ker(H_k - k\mu(\frac{1}{2})) = 2^k.$$

Since $2^k \rightarrow \infty$, the requested dimensions are arbitrarily large. $\square$

4 Verification and scope

- **Hypotheses.** The annulus stays a positive distance from the origin, so the half-flux potential is smooth and bounded. Finite products are bounded, connected, and satisfy the cone property.
- **Exactness of the building block.** The quantitative inequality (2) excludes every angular mode except $\ell = 0, 1$; Sturm–Liouville simplicity excludes hidden radial multiplicity.
- **Exactness of the product count.** The tensor-sum spectrum shows that no excited factor can contribute at the bottom. The multiplicity is therefore exactly 2^k , not merely at least 2^k .

- **Scope.** The construction lets the ambient dimension grow as $2k$. This answers the literal source request, which places no fixed-dimension restriction. It does not claim the stronger assertion that ground multiplicity is unbounded while a particular n , such as $n = 2$, is held fixed.

5 Bounded novelty check

The run’s registry, solution, attempt, and proof-gap indexes were searched by arXiv id, title, and the phrases ”ground state manifold” and ”magnetic ground-state multiplicity”; no duplicate answer was found. A local full-source search found later citations including the fractional magnetic extension arXiv:2309.06919, but no discussion resolving this multiplicity question. Web searches used the exact paper title together with ”arbitrarily large”, ”ground state manifold”, ”multiplicity”, ”tensor product”, and ”product of annuli”, and also searched arXiv for half-flux annular magnetic Neumann ground states. No prior statement of this product construction was found in that bounded search. This is not a definitive specialist-literature certification.

6 Human-review recommendation

Verify (i) that the source question is intended literally rather than with an unstated fixed- n requirement, (ii) the product-domain cone-property claim, and (iii) the tensor-sum identification for magnetic Neumann forms. A specialist citation search should precede any novelty claim beyond ”candidate full resolution”.

References

- [1] E. H. Lieb, R. Seiringer, and J. Yngvason, *Poincaré inequalities in punctured domains*, Ann. of Math. (2) **158** (2003), 1067–1080; arXiv:math/0205088.